

$$A = \begin{pmatrix} \hat{n}_0 & \hat{n}_1 \\ 0 & \hat{n}_0 \end{pmatrix}; v_0 = \begin{pmatrix} 1 & 0 \end{pmatrix}; v_1 = \begin{pmatrix} 0 & 1 \end{pmatrix}; v_2 = \begin{pmatrix} 1 & 1 \end{pmatrix};$$

Row and Column Constraint,

$$1 = \sum_i n_i =$$

$$\begin{aligned} & \hat{n}_1 \otimes \hat{n}_0 \otimes \hat{n}_0 \otimes \hat{n}_0 + \hat{n}_0 \otimes \hat{n}_1 \otimes \hat{n}_0 \otimes \hat{n}_0 + \hat{n}_0 \otimes \hat{n}_0 \otimes \hat{n}_1 \otimes \hat{n}_0 + \hat{n}_0 \otimes \hat{n}_0 \otimes \hat{n}_0 \otimes \hat{n}_1 = v_0 \cdot A \cdot A \cdot A \cdot A \cdot v_1^T; \\ & = \begin{pmatrix} 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} \hat{n}_0 & \hat{n}_1 \\ 0 & \hat{n}_0 \end{pmatrix} \cdot \begin{pmatrix} \hat{n}_0 & \hat{n}_1 \\ 0 & \hat{n}_0 \end{pmatrix} \cdot \begin{pmatrix} \hat{n}_0 & \hat{n}_1 \\ 0 & \hat{n}_0 \end{pmatrix} \cdot \begin{pmatrix} \hat{n}_0 & \hat{n}_1 \\ 0 & \hat{n}_0 \end{pmatrix} \cdot \begin{pmatrix} 0 \\ 1 \end{pmatrix}; \end{aligned}$$

Diagonal and Anti - diagonal Constraint,

$$1 \geq \sum_i n_i =$$

$$\begin{aligned} & \hat{n}_0 \otimes \hat{n}_0 \otimes \hat{n}_0 \otimes \hat{n}_0 + \hat{n}_1 \otimes \hat{n}_0 \otimes \hat{n}_0 \otimes \hat{n}_0 + \hat{n}_0 \otimes \hat{n}_1 \otimes \hat{n}_0 \otimes \hat{n}_0 + \hat{n}_0 \otimes \hat{n}_0 \otimes \hat{n}_1 \otimes \hat{n}_0 + \hat{n}_0 \otimes \hat{n}_0 \otimes \hat{n}_0 \otimes \hat{n}_1; \\ & = v_0 \cdot A \cdot A \cdot A \cdot A \cdot v_2^T = \begin{pmatrix} 1 & 0 \end{pmatrix} \cdot \begin{pmatrix} \hat{n}_0 & \hat{n}_1 \\ 0 & \hat{n}_0 \end{pmatrix} \cdot \begin{pmatrix} \hat{n}_0 & \hat{n}_1 \\ 0 & \hat{n}_0 \end{pmatrix} \cdot \begin{pmatrix} \hat{n}_0 & \hat{n}_1 \\ 0 & \hat{n}_0 \end{pmatrix} \cdot \begin{pmatrix} \hat{n}_0 & \hat{n}_1 \\ 0 & \hat{n}_0 \end{pmatrix} \cdot \begin{pmatrix} 1 \\ 1 \end{pmatrix}; \end{aligned}$$

$$A_{\theta,0}^{\alpha\beta} = A_{1,1}^{\alpha\beta} = \hat{n}_0 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}; A_{\theta,1}^{\alpha\beta} = \hat{n}_1 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix};$$

$$T_{ijkl,i'j'k'l'}^{\alpha\alpha'} = \sum_{\beta\gamma\beta'} A_{i,i'}^{\alpha\beta} A_{j,j'}^{\beta\gamma} A_{k,k'}^{\gamma\beta'} A_{l,l'}^{\beta'\alpha'};$$

$$|\Psi\rangle = T_{ijkl,i'j'k'l'}^{\alpha\alpha'} (|\theta\rangle + |1\rangle)^{\otimes N^2};$$

$$\hat{n}_0 |\theta\rangle = |\theta\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \hat{n}_0 |1\rangle = 0;$$

$$\hat{n}_1 |1\rangle = |1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}, \hat{n}_1 |\theta\rangle = 0;$$

$$Z = \langle \Psi | (|\theta\rangle + |1\rangle)^{\otimes N^2}$$